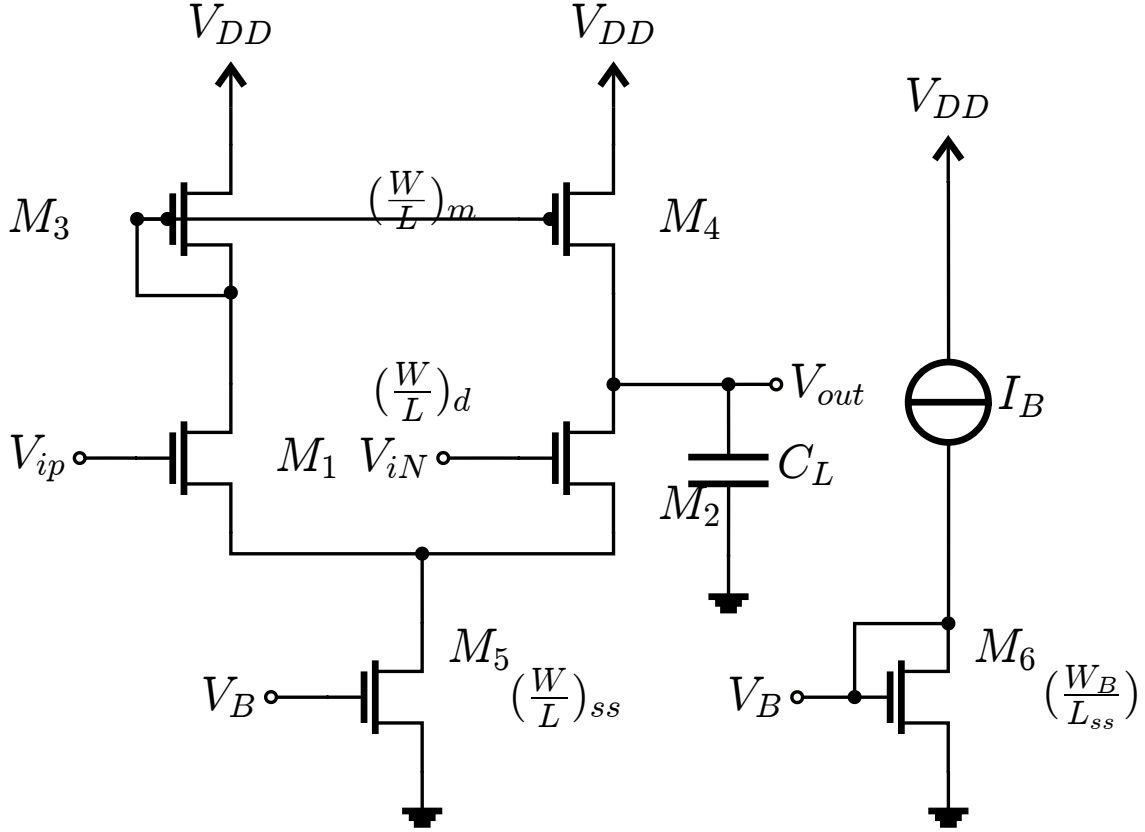


OTA5T SPECS FORMULA DERIVATIONS

5T OTA Symbolic Equations & Parameter Mapping



1. Initial Mathematical Relations

The foundational equations governing the 5T OTA circuit are defined as follows:

- **Input Common-Mode Minimum:**

$$V_{iCMmin} = V_{GS2} + V_5^*$$

- **Voltage Gain (Differential):**

$$A_v = -g_{m2}(r_{o2} \parallel r_{o4})$$

- **Input Common-Mode Maximum:**

$$V_{iCMmax} = V_{DD} + V_{t1} - V_{t3}$$

- **Unity Gain Frequency:**

$$UGF = \frac{g_{m2}}{2\pi C_L}$$

- **Common-Mode Rejection Ratio:**

$$CMRR = \frac{A_{vd}}{A_{vCM}}$$

- **Common-Mode Gain:**

$$A_{vCM} = \frac{1}{2g_{m3}r_{o5}}$$

- **Expanded CMRR:**

$$CMRR = 2g_{m2}g_{m3}(r_{o2} \parallel r_{o4})r_{o5}$$

- **Phase Margin:**

$$PM = \arctan \left(\frac{S_{pnd}}{S_u} \right)$$

- Non-dominant pole:

$$S_{pnd} = \frac{-g_{m3}}{C_{mir}}$$

- Unity gain frequency (radians):

$$S_u = -2\pi \cdot UGF = \frac{-g_{m2}}{C_L}$$

- **Power Consumption:**

$$P_w = V_{DD} \cdot (I_B + I_{ss})$$

2. Phase Margin (PM) Derivation

Expanding the Phase Margin equation based on the device parameters:

$$PM = \arctan \left(\frac{g_{m3}}{g_{m2}} \cdot \frac{C_L}{C_{mir}} \right)$$

Given the transconductances and the fact that $I_{D2} = I_{D3} = \frac{I_{ss}}{2}$:

$$g_{m2} = \sqrt{2\mu_n C_{ox} \left(\frac{W}{L}\right)_2 I_{D2}}$$

$$g_{m3} = \sqrt{2\mu_p C_{ox} \left(\frac{W}{L}\right)_3 I_{D3}}$$

Mapping the dimensions to the layout variables ($W_2 = W_d, L_2 = L_d, W_3 = W_m, L_3 = L_m$) and approximating the mirror parasitic capacitance $C_{mir} \approx W_3 L_3 C_{par}$:

$$PM \approx \arctan \left(\sqrt{\frac{\mu_p}{\mu_n}} \sqrt{\frac{W_3 L_2}{W_2 L_3}} \cdot \frac{C_L}{W_3 L_3 C_{par}} \right)$$

$$PM \approx \arctan \left(\sqrt{\frac{\mu_p}{\mu_n}} \sqrt{\frac{L_d}{W_d W_m L_m^2}} \cdot \frac{C_L}{C_{par}} \right)$$

Trade-off Deductions for PM:

There are two primary ways to increase Phase Margin:

1. **Increase C_L :** Adding an extra capacitor in parallel increases PM but directly causes UGF to drop ($UGF = \frac{g_{m2}}{2\pi C_L}$).
2. **Lower g_{m2} :** Since I_{D2} is held constant by the tail ($I_{D2} = \frac{I_{ss}}{2}$), decreasing g_{m2} requires lowering W_d . This increases PM but also causes UGF to drop. Conclusion: Any method to increase PM will likely degrade UGF.

3. Unity Gain Frequency (UGF) Derivation

$$UGF = \frac{g_{m2}}{2\pi C_L} = \frac{g_{md}}{2\pi C_L} = \frac{\sqrt{2\mu_n C_{ox} \left(\frac{W}{L}\right)_d \left(\frac{I_{ss}}{2}\right)}}{2\pi C_L}$$

Substituting the tail current equation $I_{ss} = \left(\frac{W_{ss}}{W_B}\right) I_B$:

$$UGF = \frac{\sqrt{\mu_n C_{ox} \left(\frac{W_d}{L_d}\right) \left(\frac{W_{ss}}{W_B}\right) I_B}}{2\pi C_L}$$

Trade-off Deductions for UGF:

1. **Increase W_d :** Increases g_{m2} and UGF , but decreases PM . (Note: Decreasing L_d is not considered a viable option here because it is the input capacitance of the next stage).

2. **Increase W_{ss} :** V_{GS6} and V_5^* remain constant. I_{D5} (I_{ss}) increases, causing UGF to increase, but it burns more Power ($P_w \uparrow$).
3. **Lower W_B :** Increases V_5^* ($I_B \approx \frac{1}{2} \mu C_{ox} \left(\frac{W_B}{L_{ss}} \right) (V_5^*)^2$). This pushes V_{iCMmin} up (worsening CMIR) but increases UGF .

4. Voltage Gain (A_v) Derivation

$$A_v = g_{m2}(r_{o2} \parallel r_{o4})$$

$$r_{o2} = \frac{V_{An}L_2}{(I_{ss}/2)}, \quad r_{o4} = \frac{V_{Ap}L_4}{(I_{ss}/2)}$$

$$A_v = \sqrt{2\mu_n C_{ox} \left(\frac{W}{L} \right)_2 \left(\frac{I_{ss}}{2} \right)} \cdot \frac{1}{(I_{ss}/2)} (V_{An}L_2 \parallel V_{Ap}L_4)$$

$$A_v = 2\sqrt{\frac{\mu_n C_{ox} (W/L)_2}{I_{ss}}} (V_{An}L_2 \parallel V_{Ap}L_4)$$

Substituting mapping variables and assuming $V_{An} \approx V_{Ap} = V_A$:

$$A_v = 2V_A \sqrt{\frac{\mu_n C_{ox} W_d W_B}{W_{ss} I_B}} \left(\frac{L_m \sqrt{L_d}}{L_d + L_m} \right)$$

Trade-off Deductions for Gain:

1. **Increase W_d :** Increases UGF and A_v .
2. **Increase W_B :** Lowers I_{ss} , which increases A_v and decreases Power, but drops UGF .
3. **Lower W_{ss} :** Lowers I_{ss} , which increases A_v and decreases Power, but drops UGF .
4. **Increase L_m :** Increases r_{o4} and A_v , but raises V_m^* which lowers V_{iCMmax} .

5. Common-Mode Rejection Ratio (CMRR) Derivation

To find CMRR, we first evaluate the common-mode gain (A_{vCM}):

$$A_{vCM} = \frac{1}{2g_{m3}r_{o5}} = \frac{\sqrt{W_{ss}I_B}\sqrt{L_m}}{2V_{An}\sqrt{\mu_p C_{ox} L_{ss}}\sqrt{W_m W_B}}$$

Since $CMRR = A_{vd} \cdot \frac{1}{A_{vCM}}$, the proportionality becomes:

$$CMRR \propto \frac{\sqrt{W_d} W_B L_{ss}}{W_{ss} I_B} \left(\frac{\sqrt{L_m L_d}}{L_m + L_d} \right)$$

Trade-off Deductions for CMRR:

1. **Decrease W_{ss} :** Lowers I_{ss} , raises r_{o5} , lowers A_{vCM} , increases $CMRR$.
2. **Increase W_B :** Lowers I_{ss} , raises r_{o5} , lowers A_{vCM} , increases $CMRR$.
3. **Increase W_m :** Lowers A_{vCM} , increases $CMRR$.
4. **Increase L_{ss} (Most Used):** Raises r_{o5} , lowers A_{vCM} , increases $CMRR$ without affecting A_{vd} .

6. Power Consumption (P_w)

$$P_w = (I_B + I_{ss})V_{DD} = \left(1 + \frac{W_{ss}}{W_B}\right) I_B V_{DD}$$

- To decrease power: Decrease W_{ss} (lowers UGF , raises A_v , raises $CMRR$) or Increase W_B (lowers UGF , raises A_v , raises $CMRR$).

7. Input Common-Mode Range (CMIR)

Minimum CMIR (V_{iCMmin}):

$$V_{iCMmin} = V_{GS2} + V_5^* \approx V_2^* + V_5^* + V_{t2}$$

$$V_{iCMmin} \approx \sqrt{\frac{I_B}{\mu_n C_{ox} W_B}} \left(\sqrt{\frac{L_d W_{ss}}{W_d}} + \sqrt{L_{ss}} \right) + V_{t2}$$

- To lower V_{iCMmin} (improve range): Increase W_B , increase W_d , or decrease W_{ss} .

Maximum CMIR (V_{iCMmax}):

$$V_{iCMmax} = V_{DD} + V_{t1} - V_{SG3} = V_{DD} + V_{t1} - V_{t3} - V_m^*$$

$$V_m^* = \frac{1}{\sqrt{\mu_p C_{ox}}} \sqrt{\frac{W_{ss} L_m I_B}{W_B W_m}}$$

$$V_{iCMmax} = V_{DD} + V_{t1} - V_{t3} - \frac{1}{\sqrt{\mu_p C_{ox}}} \sqrt{\frac{W_{ss} L_m I_B}{W_B W_m}}$$

- To raise V_{iCMmax} (improve range): Increase W_B or increase W_m .

8. Full Symbolic Equations Summary

$$PM \approx \arctan \left(\sqrt{\frac{\mu_p}{\mu_n}} \sqrt{\frac{L_d}{W_d W_m L_m^2}} \cdot \frac{C_L}{C_{par}} \right)$$

$$UGF = \frac{\sqrt{\mu_n C_{ox} \left(\frac{W_d}{L_d} \right) \left(\frac{W_{ss}}{W_B} \right) I_B}}{2\pi C_L}$$

$$A_v = 2V_A \sqrt{\frac{\mu_n C_{ox} W_d W_B}{W_{ss} I_B}} \left(\frac{L_m \sqrt{L_d}}{L_d + L_m} \right)$$

$$CMRR \propto \frac{\sqrt{W_d} W_B L_{ss}}{W_{ss} I_B} \left(\frac{\sqrt{L_m L_d}}{L_m + L_d} \right)$$

$$P_w = \left(1 + \frac{W_{ss}}{W_B} \right) I_B V_{DD}$$

$$V_{iCMmin} \propto \sqrt{\frac{I_B}{\mu_n C_{ox} W_B}} \left(\sqrt{\frac{L_d W_{ss}}{W_d}} + \sqrt{L_{ss}} \right)$$

$$V_{iCMmax} = V_{DD} + V_{tn} - V_{tp} - \frac{1}{\sqrt{\mu_p C_{ox}}} \sqrt{\frac{W_{ss} L_m I_B}{W_B W_m}}$$

9. Map from Parameter to Target Effect

- $W_d \uparrow$ **(Increase):**
 - $PM \downarrow$
 - $UGF \uparrow$
 - $A_v \uparrow$
 - $CMRR \uparrow$
 - P_w (Constant)
 - $V_{iCMmin} \downarrow$ (Improves)
 - V_{iCMmax} (Constant)
- $W_{ss} \uparrow$ **(Increase):**
 - PM (Almost Constant)
 - $UGF \uparrow$
 - $A_v \downarrow$

- $CMRR \downarrow$
 - $P_w \uparrow$
 - $V_{iCMmin} \uparrow$ (Degrades)
 - V_{iCMmax} (Constant)
- $L_{ss} \uparrow$ **(Increase):**
 - PM (Constant)
 - UGF (Constant)
 - A_v (Constant)
 - $CMRR \uparrow$
 - P_w (Constant)
 - $V_{iCMmin} \uparrow$ (Degrades)
 - V_{iCMmax} (Constant)
- $W_B \uparrow$ **(Increase):**
 - PM (Almost Constant)
 - $UGF \downarrow$
 - $A_v \uparrow$
 - $CMRR \uparrow$
 - $P_w \downarrow$
 - $V_{iCMmin} \downarrow$ (Improves)
 - $V_{iCMmax} \uparrow$ (Improves)
- $W_m \uparrow$ **(Increase):**
 - $PM \downarrow$ (Decreases slightly)
 - UGF (Constant)
 - A_v (Constant)
 - $CMRR$ (Constant)

- P_w (Constant)
- V_{iCMmin} (Constant)
- $V_{iCMmax} \uparrow$ (Improves)
- $L_m \uparrow$ (Increase):
 - $PM \downarrow$ (Decreases a lot)
 - UGF (Constant)
 - $A_v \uparrow$
 - $CMRR \downarrow$ (Decreases slightly)
 - V_{iCMmin} (Constant)
 - $V_{iCMmax} \downarrow$ (Degrades)

Heuristic Trade-off Rules (The Designer's Map)

To decide *which* parameter to modify, do not guess or attempt complex calculus. Use the following established heuristic rules. These rules act as a lookup table showing exactly how modifying a specific transistor dimension affects the circuit specifications, including whether that change is beneficial (✓) or detrimental (✗) to our target constraints.

The "INCREASING" Map (Multiplying by 1.05)

- **Increasing Wd:**
 - Increases: UGF (✓), A_v (✓), $CMRR$ (✓)
 - Decreases: V_{icmmin} (✓), PM (✗)
 - Constant: P_w , V_{icmmax}
- **Increasing Wss:**
 - Increases: UGF (✓), P_w (✗), V_{icmmin} (✗)
 - Decreases: A_v (✗), $CMRR$ (✗)
 - Constant: PM , V_{icmmax}
- **Increasing Lss:**

- Increases: CMRR (✓), Vicmmin (✗)
 - Decreases: Nothing
 - Constant: PM, UGF, Av, Pw, Vicmmax
- **Increasing WB:**
 - Increases: Av (✓), CMRR (✓), Vicmmax (✓)
 - Decreases: UGF (✗), Pw (✓), Vicmmin (✓)
 - Constant: PM
- **Increasing Wm:**
 - Increases: Vicmmax (✓)
 - Decreases: PM (✗ slightly)
 - Constant: UGF, Av, CMRR, Pw, Vicmmin
- **Increasing Lm:**
 - Increases: Av (✓)
 - Decreases: PM (✗ significantly), CMRR (✗), Vicmmax (✗)
 - Constant: UGF, Pw, Vicmmin

The "DECREASING" Map (Multiplying by 0.95)

- **Decreasing Wd:**
 - Decreases: UGF (✗), Av (✗), CMRR (✗)
 - Increases: Vicmmin (✗), PM (✓)
 - Constant: Pw, Vicmmax
- **Decreasing Wss:**
 - Decreases: UGF (✗), Pw (✓), Vicmmin (✓)
 - Increases: Av (✓), CMRR (✓)

- Constant: PM, Vicmmax
- **Decreasing Lss:**
 - Decreases: CMRR (✗), Vicmmin (✓)
 - Increases: Nothing
 - Constant: PM, UGF, Av, Pw, Vicmmax
- **Decreasing WB:**
 - Decreases: Av (✗), CMRR (✗), Vicmmax (✗)
 - Increases: UGF (✓), Pw (✗), Vicmmin (✗)
 - Constant: PM
- **Decreasing Wm:**
 - Decreases: Vicmmax (✗)
 - Increases: PM (✓ slightly)
 - Constant: UGF, Av, CMRR, Pw, Vicmmin
- **Decreasing Lm:**
 - Decreases: Av (✗)
 - Increases: PM (✓ significantly), CMRR (✓), Vicmmax (✓)
 - Constant: UGF, Pw, Vicmmin

Agent Instruction: Analyze the stretched target. Based strictly on the mapping rules above, logically deduce **one** primary physical parameter (from Wd, Wm, Lm, Wss, Lss, WB) that should be modified to improve the target spec without breaking the others.

- Search the maps for an action where the target you want to change is marked with a (✓).
- Verify that you have enough margin on the specs marked with a (✗) before selecting that action.

- Select a backup parameter and direction in case your primary choice fails to improve the spec during the step-search.

Step 2: Heuristic Trade-off Rules (The Designer's Map)

The circuit specifications [Av, UGF, CMRR, PM, Pw, Av_VICM_MIN, Av_VICM_MAX] are direct functions of the physical parameters (Wd, Ld, Wm, Lm, Wss, Lss, WB) .

To decide *which* parameter to modify, do not guess or attempt complex calculus. Use the following established heuristic rules. These rules act as a lookup table showing exactly how modifying a specific transistor dimension affects the circuit specifications, including whether that change is beneficial (✓) or detrimental (✗) to our target constraints.

The "INCREASING" Map (Multiplying by 1.05)

- **Increasing Wd:**
 - Increases: UGF (✓), Av (✓), CMRR (✓), Av_VICM_MIN (✓)
 - Decreases: PM (✗)
 - Constant: Pw, Av_VICM_MAX
- **Increasing Wss:**
 - Increases: UGF (✓), Pw (✗)
 - Decreases: Av (✗), CMRR (✗), Av_VICM_MIN (✗)
 - Constant: PM, Av_VICM_MAX
- **Increasing Lss:**
 - Increases: CMRR (✓)
 - Decreases: Av_VICM_MIN (✗)
 - Constant: PM, UGF, Av, Pw, Av_VICM_MAX
- **Increasing WB:**
 - Increases: Av (✓), CMRR (✓), Av_VICM_MAX (✓), Av_VICM_MIN (✓)

- Decreases: UGF (✗), Pw (✓)
- Constant: PM
- **Increasing Wm:**
 - Increases: Av_VICM_MAX (✓)
 - Decreases: PM (✗ slightly)
 - Constant: UGF, Av, CMRR, Pw, Av_VICM_MIN
- **Increasing Lm:**
 - Increases: Av (✓)
 - Decreases: PM (✗ significantly), CMRR (✗), Av_VICM_MAX (✗)
 - Constant: UGF, Pw, Av_VICM_MIN

The "DECREASING" Map (Multiplying by 0.95)

- **Decreasing Wd:**
 - Decreases: UGF (✗), Av (✗), CMRR (✗), Av_VICM_MIN (✗)
 - Increases: PM (✓)
 - Constant: Pw, Av_VICM_MAX
- **Decreasing Wss:**
 - Decreases: UGF (✗), Pw (✓)
 - Increases: Av (✓), CMRR (✓), Av_VICM_MIN (✓)
 - Constant: PM, Av_VICM_MAX
- **Decreasing Lss:**
 - Decreases: CMRR (✗)
 - Increases: Av_VICM_MIN (✓)
 - Constant: PM, UGF, Av, Pw, Av_VICM_MAX

- **Decreasing WB:**
 - Decreases: A_v (✗), CMRR (✗), $A_{v_VICM_MAX}$ (✗), $A_{v_VICM_MIN}$ (✗)
 - Increases: UGF (✓), P_w (✗)
 - Constant: PM
- **Decreasing W_m :**
 - Decreases: $A_{v_VICM_MAX}$ (✗)
 - Increases: PM (✓ slightly)
 - Constant: UGF, A_v , CMRR, P_w , $A_{v_VICM_MIN}$
- **Decreasing L_m :**
 - Decreases: A_v (✗)
 - Increases: PM (✓ significantly), CMRR (✓), $A_{v_VICM_MAX}$ (✓)
 - Constant: UGF, P_w , $A_{v_VICM_MIN}$

Agent Instruction: Analyze the stretched target. Based strictly on the mapping rules above, logically deduce **one** primary physical parameter (from W_d , W_m , L_m , W_{ss} , L_{ss} , WB) that should be modified to improve the target spec without breaking the others.

- Search the maps for an action where the target you want to change is marked with a (✓).
- Verify that you have enough margin on the specs marked with a (✗) before selecting that action.
- Select a backup parameter and direction in case your primary choice fails to improve the spec during the step-search.

Agent Instruction: Analyze the stretched target. Based strictly on the mapping rules above, logically deduce **one** physical parameter (from W_d , L_d , W_m , L_m , W_{ss} , L_{ss} , W_B) that should be modified to improve the target spec without breaking the others.

- Look for a parameter where the target you want to change has a (✓), in other words search the maps for an action where the target you want to change is marked with a (✓).
- Verify that you have enough margin on the specs marked with a (✗) before making the change.
- Determine whether it needs to be increased or decreased (e.g., If a rule says "Increasing WB decreases Pw (✓)", then to lower power, you must increase WB by a small step aka step WB upwards).
- (Contingency Plan) Select a backup parameter and direction in case your primary choice fails to improve the spec during the step-search.

Agent Instruction for Building the Execution Plan:

You must build a multi-step `execution_plan` list containing all parameters that help your target. Order them strictly by safety margin:

1. **Primary Choice (Safest):** Find a parameter that has a (✓) for your target spec, and has NO detrimental effects (✗) on any other specs. If that doesn't exist, pick one where the (✗) affects a specification where we currently have a massive passing margin.
2. **Contingency 1 (Moderate Risk):** A parameter that improves the target but has a (✗) on a specification that is somewhat close to its failure limit.
3. **Contingency 2 (High Risk):** A parameter that improves the target but negatively impacts a very sensitive specification (like PM).

Example of Execution Plan Logic:

If you want to increase UGF, and your current PM is dangerously close to 60° , you should put W_{ss} as your Primary Choice (because increasing W_{ss} doesn't hurt PM), and put W_d as a contingency (because increasing W_d lowers PM).